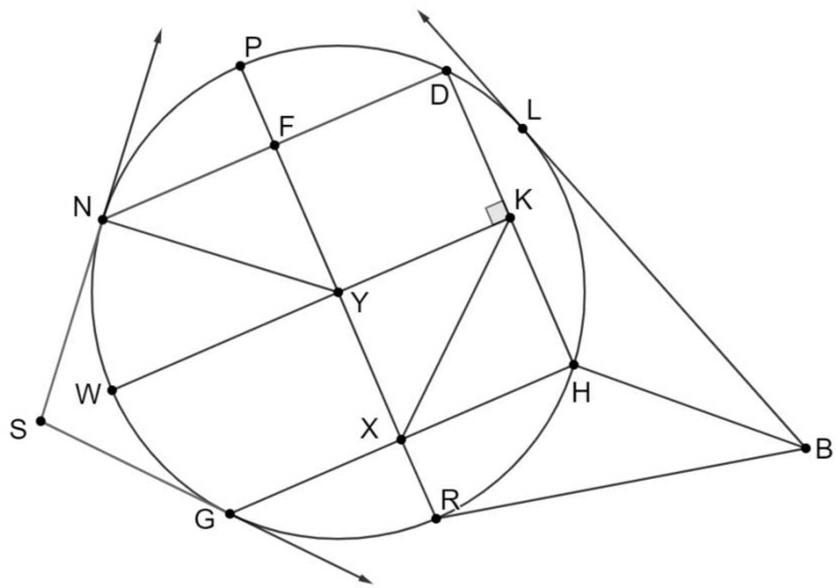
Geometry
Unit 12
Calculator

Name____________________
Date_________________
Period__________

1. Circle Y is shown where points L, H, R, G, W, N, P, and D are on the circle. $\overrightarrow{SN}$ is tangent to the circle at point N, $\overrightarrow{BL}$ is tangent to the circle at point L, and $\overrightarrow{SG}$ is tangent to the circle at point G. $\overline{WK}$ is a perpendicular bisector of $\overline{DH}$ and $\overline{FY} \cong \overline{YX}$.

Select all of the statements that are true based on the given information.

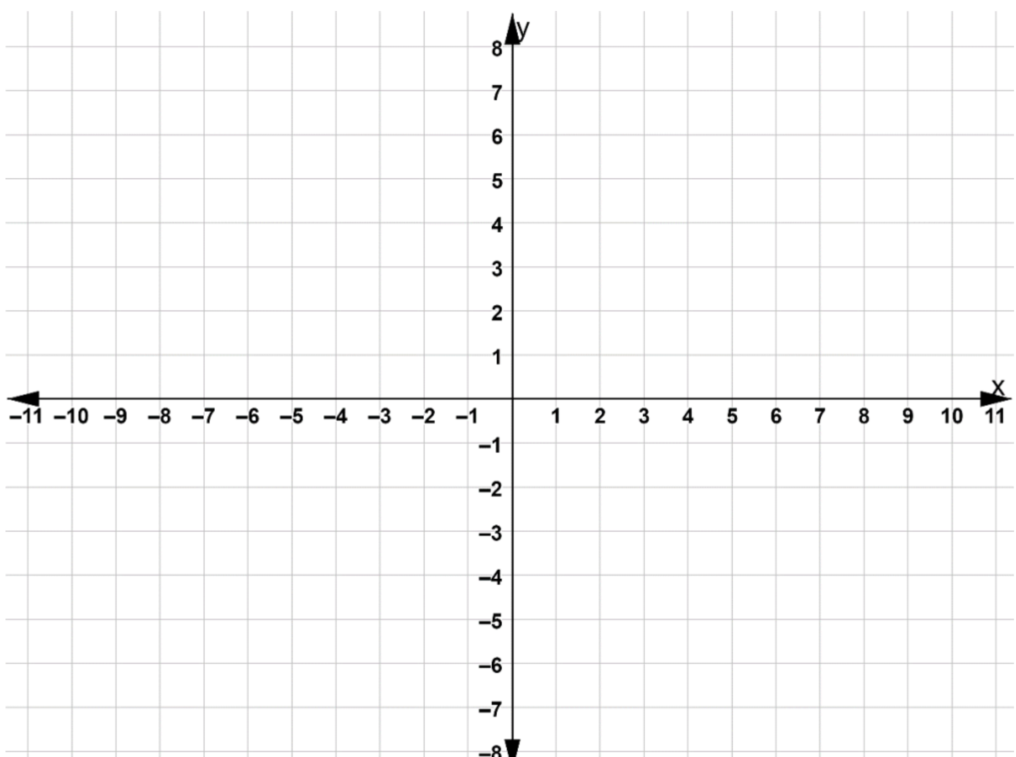
- ☐ $\overline{ND} \cong \overline{WK}$
- ☐ $\overline{NY} \cong \overline{XK}$
- ☐ $\overline{BR} \cong \overline{BL}$
- ☐ $\overline{DK} \cong \overline{KH}$
- ☐ $\overline{NS} \cong \overline{SG}$
- ☐ $\overline{GH} \cong \overline{ND}$



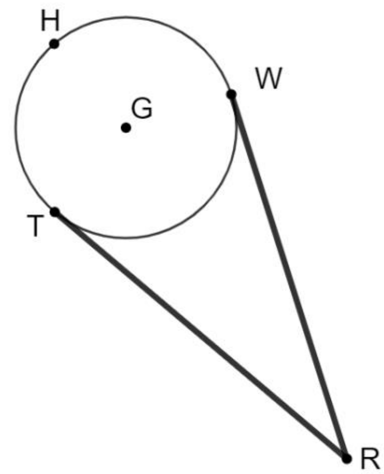
2. An equation of a circle is shown.

$(x + 6)^2 + (y - 3)^2 = 4$

Graph the circle.



3. Hydropower plants produce electricity by transforming energy from water wheels to electricity. Rural areas of poor countries often do not have access to electricity, so using a small stream with a modern water wheel can give the area electricity. The diagram shows circle G which represents a modern water wheel. A belt extends from point R to point T, then it lays on the circumference of circle G passing through points H and W, connecting again at point R.

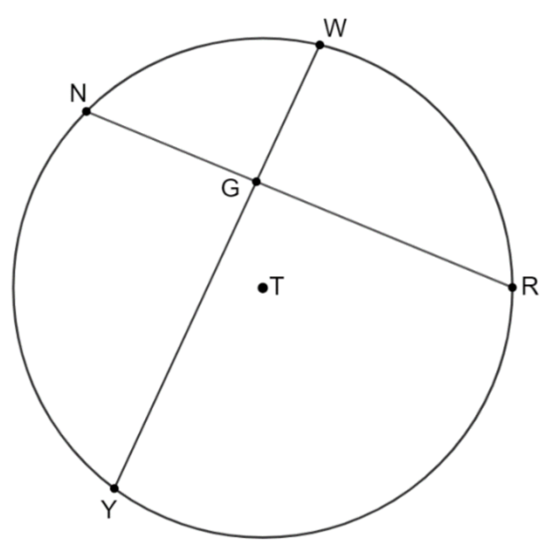


$WR = 48.5$ centimeters (cm) and
$TR = (7k - 19.4)$ cm

Determine the value of k.

$\boldsymbol{k} =$	

4. Circle T is shown with points N, W, R, and Y on the circle. $\overline{RN}$ intersects $\overline{YW}$ at point G. $WG = 7.5$, $NG = 9$, $GR = 14.25$, and $YG = x$.

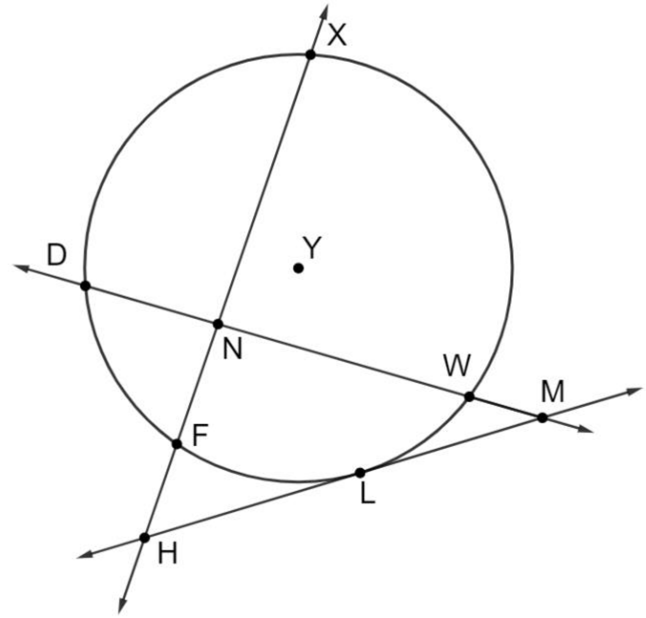


Determine the value of x.

$\boldsymbol{x} =$	

5. Circle Y is shown with points X, W, L, F, and D on the circle. $\overleftrightarrow{HM}$ is tangent to circle Y at point L. $\overline{DW}$ intersects $\overline{FX}$ at point N. $FN = 63$, $NX = 102$, $HF = 55$, and $HL = b$.

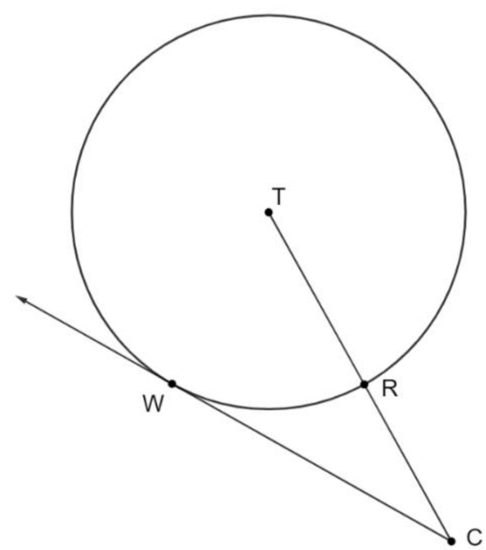
Determine the value of b.



$b =$	

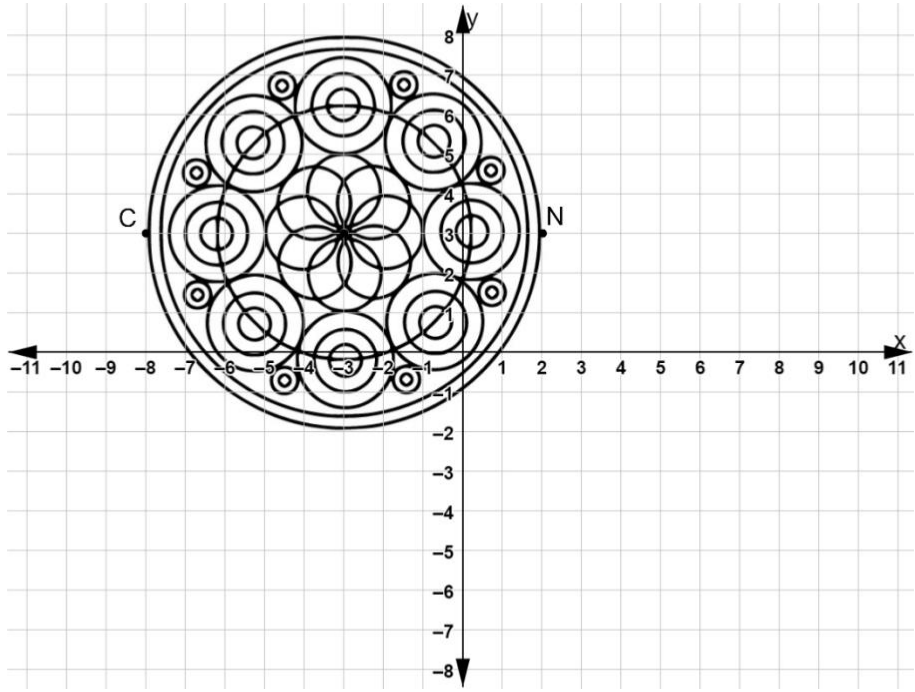
6. Circle T is shown with points R and W on the circle. $\overrightarrow{CW}$ is tangent to circle T at point W. $TR = x$, $RC = 5$, and $WC = 9$.

Determine the length of circle T's diameter.



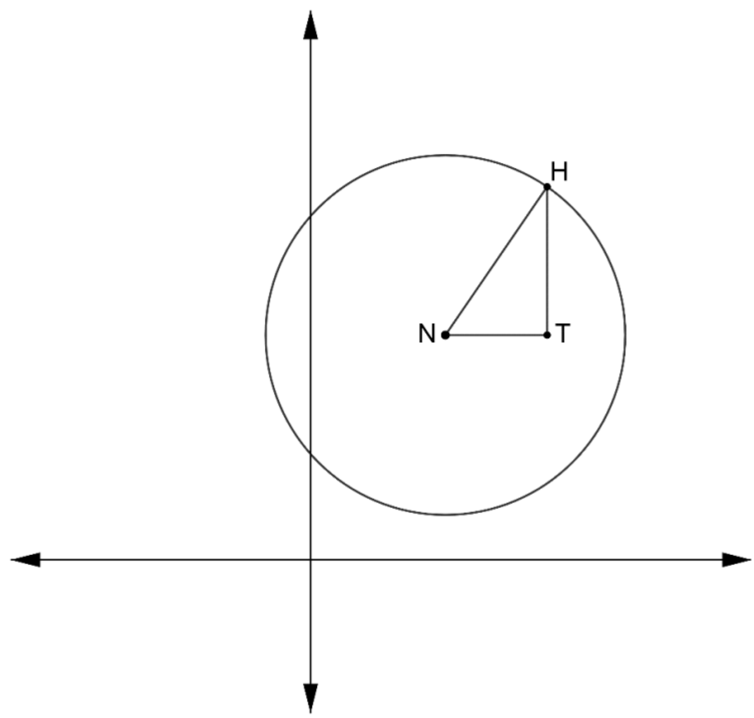
Diameter	

7. A mandala is a geometric design used in the Hindu culture. A graphic artist uses a drawing program to create a mandala. A diameter of the largest circle has endpoints C $(-8,3)$ and N $(2,3)$. Write the equation of the circle that passes through the points C and N.



Equation	

8. Circle N is shown where point H is on the circle. Point N is located at $(5, 8)$, point H is located at (x, y) and $m\angle HTN = 90°$.



a. Complete the table.

Segment	Length
$\overline{HT}$	
$\overline{NT}$	

b. The segment $\overline{NH}$ represents the radius of the circle. Using r to represent the length of $\overline{NH}$ and the lengths in the table, write an equation that shows the relationships of the side lengths of ΔHTN.

Equation	

c. Complete the statements.

The equation that shows the relationships of the side lengths of ΔHTN also represents the equation of circle N because point

- ○ H
- ○ T
- ○ N

is a representative point of

- ○ ΔHTN
- ○ circle N

and can be placed anywhere on the

- ○ circle.
- ○ triangle.

Any

- ○ circle
- ○ right triangle

where the

- ○ radius
- ○ hypotenuse

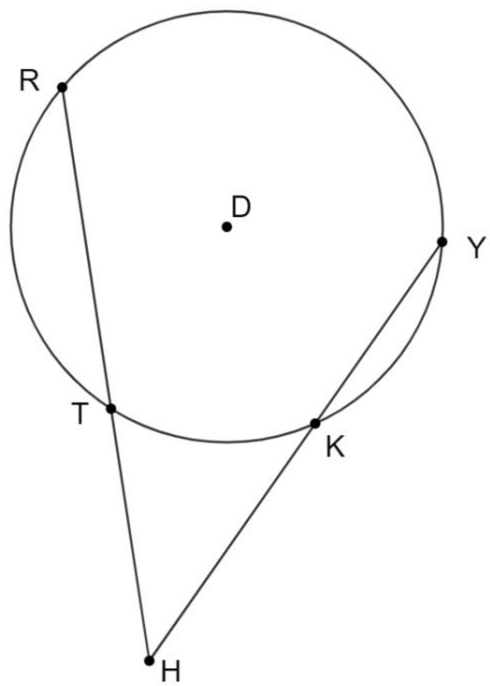
is the

- ○ radius of the circle
- ○ hypotenuse of the triangle

will result in the same side lengths.

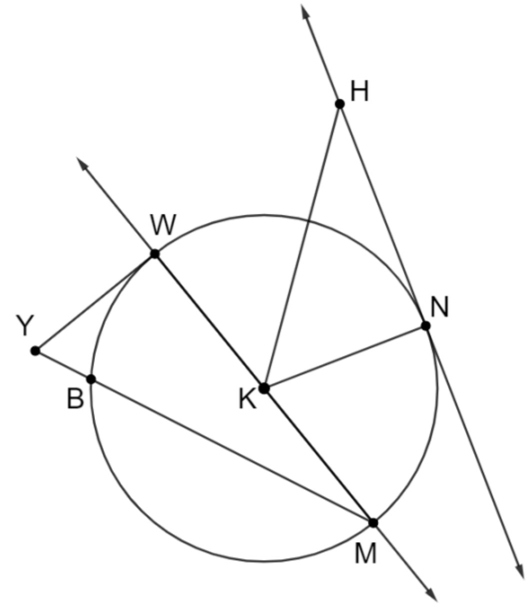
9. Circle D is shown with points R, T, K, and Y on the circle. $HY = 3x + 1$, $TH = 2x - 4$, $RH = 32$, and $HK = 16$.

Determine the value of x.



$x =$	

10. Circle K is shown where points N, M, B, and W are on the circle. $\overline{YW}$ is tangent to circle K at point W and $\overleftrightarrow{HN}$ is tangent to circle K at point N. $WY = 4$, $NK = 4.5$, and $HN = 6$.



a. Determine the length of $\overline{KH}$.

$KH =$	

b. Determine the length of $\overline{YB}$. If necessary, round your answer to the nearest tenth.

$YB =$	

11. An equation of a circle is shown.

$$x^2 + y^2 - 8x + 14y - 79 = 0$$

Complete the table of key features.

<table>
<tr><th>Center-Radius form</th><th>Center</th><th>Diameter</th></tr>
<tr><td rowspan="3">work

Equation | |</td><td></td><td></td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td></td><td></td></tr>
</table>